

BrightLearn Data Analytics

Research Assignment 1

Due Date: 14 September 2026

Section A: Data Fundamentals

1. A database is a digital storage location for storing, managing and securing organized collections of data.

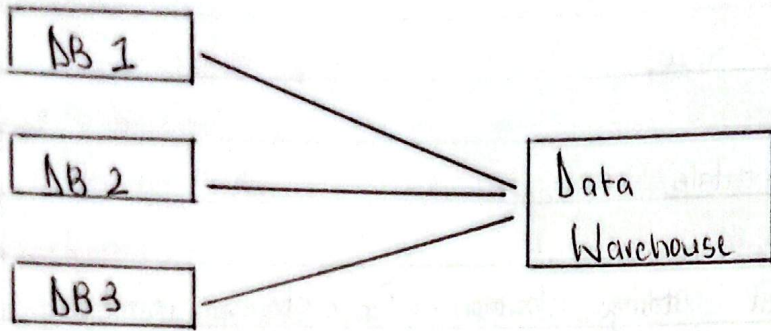
- Databases are essential in the modern world because they allow us to organize, and protect massive amounts of data so that businesses can be able to revisit the data in cases whereby historical data needs to be revisited. And most importantly it allows businesses to analyse the data and be able to see how the business is doing, to be able to spot areas where improvement is needed and also to be able to make predictions about the future of the business based on certain trends spotted on the data.

- In sales, databases are used to track customer interactions in CRM software in order to manage product inventory and pricing.

- In banking apps, databases act as a secure back up system should the banking app crash.

2. A data warehouse is a centralized storage system designed to collect, combine and analyse large volumes of ~~current~~ ^{current} and historical data from multiple sources viz. multiple databases.

Eg



- A data warehouse is different from a regular database in that it is optimized for OLAP (Online Analytical Processing). It is built to store historical data and to run heavy complex queries. Meanwhile a regular database is optimized for OLTP (Online Transaction Processes). It is built to store real time data from daily transactions from users or applications as they happen.
- A company can use a data warehouse because it turns raw data from different sources into a single, high performance engine for making business decisions.
- 3. A Data Lake is a centralized digital storage location designed to store vast amounts of raw data in its original, unformatted state. A data lake accepts/stores data of any type, volume, or speed, holding it until it is needed for analysis.
- A Data Lake is different from a data warehouse in that it stores its data in a state of being raw and unprocessed, while a data warehouse does the opposite, it stores its data in a state of being processed, cleaned and structured.

- This means that a data lake stores unstructured data and a data warehouse stores structured data.

- A business would choose to use a data lake over a data warehouse when they need to store massive volumes of raw, unstructured data for advanced AI, machine learning and data sciences at very low costs.

4. A data mart is a smaller, more specific version of a data warehouse that is built to handle reporting and analytical needs of a single department or business unit.

- It relates to a data warehouse in that it is a smaller component of a central data warehouse, focusing on just one subject instead of the entire enterprise.

- It is typically used by department analysts, business stakeholders and data engineers.

5. Structured, semi-structured and unstructured data represent the three main categories of how information is organized, stored and processed. E.g Microsoft Excel spreadsheet

- Structured data is highly organized and fits neatly into tables with rows and columns. E.g XML file.

- Semi-structured data doesn't fit into tables but has some organization through tags or labels to make it readable.

- Unstructured data has no set structure or organization at

all. E.g. a raw video recording.

- 6. Data Modelling is the process of designing a visual plan that shows how data will be organized, stored, and connected to each other within a system.
 - It is important because it serves as a universal communication tool that translates complex business requirements into a clear visual format thus bridging the gap between non-technical and technical business managers.
 - Conceptual Data Modelling - It is more like a big picture view of your data. It focuses on identifying the main entities in a business and the relationship between them, without worrying about how it will actually be built in a database.
 - Logical Data Modelling - It is the more detailed version that comes after conceptual model. It defines exactly what data attributes each entity will have, the data types, the specific rules for how entities relate to each other without worrying about which database software will be used.
- 7. Data Mining is the process of analyzing large amounts of data using statistics and machine learning to uncover hidden patterns, trends and relationships that are not obvious at first glance.
 - Businesses use data mining to turn raw operational data into actionable insights.

Two real world examples are:

Retail Product Recommendation: Online retailers analyze your past browsing habits, click history, and previous purchases, to predict what products you might want to buy next and display personalized recommendations on the homepage. This increases sales and improves the shopping experience.

Credit Card Fraud Detection: Financial institutions monitor millions of credit card transactions in real time. Data mining models flag transactions that deviate significantly from a user's normal spending habits, such as sudden overseas purchase. The bank can then instantly block the card or alert the customer to prevent fraud.

s. A Database Management System is software that sits between the users and the actual database. It lets you create, read, update, and delete data easily, while it takes care of keeping the data organized, secure, consistent, and backed up in the background.

- MySQL - a popular open-source database system that uses SQL to store and manage data. It keeps information in organized tables and makes it easy to save, find, and update large amounts of data quickly and reliably.

- PostgreSQL - is a powerful open-source database system that is known for being very stable, secure, and advanced. Like MySQL it uses tables and SQL, but it handles complex data and large amounts of information even better, which is why many big companies prefer it.

- Oracle Database - is a very powerful, commercial database system made by Oracle Corporation. It is designed for large businesses that need to handle huge amounts of data with high security, performance, and reliability.
- MongoDB - is a popular NoSQL database that doesn't store data in tables like MySQL does. Instead, it stores as flexible, JSON like documents, which makes it very fast and easy to use for modern apps where the data structure can change often.

MySQL is suitable for Web applications and commerce platforms.

PostgreSQL is suitable for Financial and geospatial applications.

Oracle Database is suitable for large-scale enterprise operations.

MongoDB is suitable for content management and real-time big data applications.

9. ETL - stands for Extract, Transform, Load. It is the process of taking data from different sources, cleaning and changing it into useful format, and then loading it into one central database or data warehouse for analysis.

Extract - The data pipeline reads and collects raw data from varied sources, such as relational databases, CRM software, cloud storage, API endpoints, or flat files (like CSVs)

Transform - Raw data is cleaned, formatted, and structured to ensure quality and consistency. This step involves removing duplicates, filtering out invalid records, converting data types, standardizing currencies or time zones, and applying business logic.

Load - Moves the processed data from the staging area into

a final destination like a data warehouse or data lake for analytics and reporting.

10. CSV - a plain text, row-oriented tabular format where each line is a record and fields are split by commas.

When to use: Use CSV for small datasets, simple data exchanges, human-readable exports, or sharing with non-technical teams.

JSON - a lightweight text format using nested objects and key-value pairs that is readable by both humans and machines.

When to use: use JSON for web APIs, event logs, and semi-structured data with nested or varying attributes.

Avoid it for heavy analytical querying due to its large file sizes and slow parsing overhead.

Apache Parquet: a binary, columnar storage format optimized for high-performance reading, analytical processing, and compression.

When to use: use Parquet for large-scale data warehousing, cloud data lakes, and analytical queries that scan specific columns rather than entire rows. Avoid it for write heavy streaming ingestion.

Apache Avro: a binary, row oriented format that stores its schema alongside the data, enabling fast serialization and updates.

When to use: use Avro for write-intensive pipelines, event-driven streaming (like Apache Kafka) and scenarios requiring robust schema evaluation.

Avoid it for final analytical reporting where columnar read performance is preferred.

iv. Artificial

Section B: Artificial Intelligence & Machine Learning

11. Artificial Intelligence (AI) is the development of computer systems to be able to perform tasks that normally need human intelligence, such as reasoning, learning, and problem-solving.

- Narrow AI is designed and trained to perform a single task or a limited set of tasks. It cannot function or learn outside its predefined scope. E.g. Apple's Siri.
- General AI is a theoretical machine that can match human intelligence across any intellectual task. It can learn, reason and adapt to completely new situations without special programming. E.g. a hypothetical robot that can independently cook a meal, write a computer programme, diagnose a medical issue and learn a new language just like a person. This kind of AI does not exist yet.

12. Machine Learning (ML) is a part of AI where we don't give the computer exact instructions for every situation. Instead, we show it a lot of data and it learns the pattern by itself, so it gets better at making predictions or decisions the more data it sees.

Machine Learning differs from traditional programming in that, in traditional programming a developer manually writes the logic, whereas Machine Learning shifts the burden of finding the rules to the computer itself.

Supervised Learning: The model learns on a labelled dataset. This means every piece of training data comes paired with the correct answers. The AI acts like a student learning from an answer key.

Unsupervised Learning: The model is given unlabelled data and must find patterns, anomalies, or structures entirely on its own. There is no teacher and no explicit target outcome.

Reinforcement Learning: The model (called an agent) learns by interacting with an environment through trial and error. It receives virtual "rewards" for correct actions and "penalties" for wrong ones, adapting its strategy over time.

13. Deep Learning is a more advanced type of Machine Learning that uses systems called neural networks, designed to work a bit like the human brain. It uses many layers to learn very complex patterns from huge amounts of data.

It is related to AI and ML in the following way:
Artificial Intelligence is the broad field of creating smart machines that can simulate human intelligence.

Machine Learning is a subset of AI where computers learn from data without being explicitly programmed.

Deep Learning: is a specialized subset of ML that uses multi-layered networks to learn from vast amounts of

unstructured data (like raw text, images, or audio) automatically.

Artificial Neural Networks (ANN) is the core software architecture that powers deep learning. It is inspired by the network of biological neurons in the human brain.

Neural networks learn by processing data, making mistakes, and adjusting their internal settings.

14. NLP is a field of AI that teaches computers to understand our everyday language. Since human language is messy with slang, jokes, and double meanings, NLP helps computers make sense of it so they can read, understand, and even talk back to us like in chatbots or translation apps.

Three everyday life examples of NLP are:

- Smartphones & voice assistants
- Autocorrect, predictive text, and smart reply.
- Email spam filtering

15. An LLM is a very powerful Deep Learning model trained on billions of texts from the internet, books etc. It doesn't really "think", it learned the patterns of language so well that its extremely good at predicting what word should come next. The ability to predict the next word is what lets it write, summarize, translate, and have conversations that sound human.

The core difference between Large Language Model and Machine Learning lies in the type of data they process and their versatility. Standard Machine Learning prefers structured data (e.g. numbers, spreadsheets, tables, databases)

Large Language Model thrives on unstructured data (e.g. books,

articles, charts, code).

The well-known LLMs are:

- ChatGPT - it functions as an all-purpose conversational assistant. People use it to brainstorm ideas, draft emails, format text, write software code and answer complex questions.
- Claude - is popular for heavy analytical work, coding, and enterprise tasks. It is highly regarded for its deep context window, meaning it can process an entire book or a massive contract in a single prompt to pull out data, analyze legal jargon, or summarise long transcripts.

16. Generative AI is a type of artificial intelligence capable of creating entirely new content, such as text, images, music, audio or video that closely resembles human created work. Unlike traditional AI, which is built to analyze data or make predictions, Generative AI learns patterns from existing data to produce original fresh outputs.

Generative AI operates through a three-step process of recognition, calculation, and creation:

1. Training on Massive Datasets: The AI is fed billions of examples of existing human creations (e.g millions of photos of dogs or vast libraries of written literature).
2. Learning the Underpinned Patterns: Instead of memorizing the files, the system breaks them down into mathematical rules, mapping how words relate to each other or how shapes and pixels form complex images.

3. Generating Outputs from Prompts: When a user inputs a prompt (e.g. "a cat wearing a spacesuit") the AI uses its learned statistical map to assemble a completely new layout, calculating word-by-word or pixel-by-pixel what should come next to fit the description.

Three Popular Generative AI Tools

- ChatGPT (by OpenAI): A text-to-text foundational model that acts as a conversational assistant. It generates human-like responses that are written, writes essays, drafts emails, and creates or debugs programming code based on natural language prompts.
- Midjourney: A text-to-image generator that translates descriptive text prompts into highly detailed, stylised artwork, digital designs, and photorealistic graphics.
- ElevenLabs: A generative audio tool that synthesizes natural-sounding speech from written text. It can clone specific voices with just a short audio sample or generate completely new, professional-grade sound effects and narration voices.

17. A training dataset is a collection of information used to teach an artificial intelligence model how to recognise patterns and make predictions.

Training Data Quality is important because:

- The AI can only be as accurate as the data it learns from.
- Clean and varied data helps the the AI handle new situations correctly.
- Fixing a broken AI model is impossible if the original training data is poor.

If training data is biased or ~~imprecise~~ incorrect it can result

in:

- Wrong answers - The AI copies mistakes and gives false or useless results.
- Unfair decisions - Biased data makes the AI treat certain groups of people unfairly.
- Real world failure - The AI might pass basic tests, but it will fail when people use it in daily life.

18. Computer Vision (CV) is a field of artificial intelligence that trains machines to interpret and understand the visual world. It gives computers the ability to extract meaningful information from digital images, videos, and visual inputs, allowing them to take action or make recommendations based on what they "see".

Machines don't "see" images like we do. To them an image is just a grid of numbers, each pixel has a number for its brightness and color.

For them to "interpret" it, we use computer vision, which is usually powered by Deep Learning:

- The model looks at the numbers and learns to find low-level patterns like edges and textures.
- It then combines those into higher-level patterns like shapes, objects, and faces in the next layers.
- Finally, it matches what it found to patterns it has seen before in training. E.g. so it says "this cluster of numbers looks like the dog pattern I learned, so this is probably a dog."

Two Real-World Cases:

Autonomous Vehicles and Advanced Driver Assistance Systems (ADAS)-

Self-driving vehicles rely heavily on computer vision as their main sensory modality.

- Healthcare and Medical Imaging Diagnostics - Computer vision acts as an automated assistant for radiologists and oncologists analyzing complex medical scans like X-rays, MRIs, and CT scans.

Section C: APIs, Tools & Developer Concepts

19. Application Programming Interface (API) - is a set of rules and protocols that allows different software applications to communicate and share data with one another.

The easiest way to understand an API is to imagine yourself sitting at a table in a restaurant:

- The Customer (The Client/User): You sit at a table looking at a menu of choices. You want to order food, but you cannot walk directly into the kitchen to cook it yourself.
- The Kitchen (The Server/Database): This is the backend system that holds all the ingredients and knows how to prepare the food.
- The waiter (The API): The waiter is the crucial link between you and the kitchen. They take your order (the request), deliver it precisely to the kitchen, and then return to your table with your food (the response).

APIs are important in modern software because they serve as the foundational plumbing of the modern web and digital economy. They are essential for several key reasons:

- Preventing "Reinventing the Wheel": Developers do not need to build every single feature from scratch. For instance, if an app needs a map interface, the developer doesn't build a mapping system or launch satellites; they simply plug into the Google Maps API.
- Security and Privacy: APIs provide controlled access. They hide internal system complexities and keep sensitive code or data locked down. When you log on e-commerce site via PayPal, a payment API handles the transaction securely, so the store never sees your raw credit and data.
- Interoperability and Automation: APIs permit completely different software systems running on different programming languages or operating systems to speak a "common language".
- Scalability and flexibility: By separating the frontend (what the user sees) from the backend data, companies can update their core systems or launch apps across multiple platforms (mobiles, web, smartwatches) using the exact same APIs.

20. An API key is a secret string of letters and numbers that a software application uses to identify itself when connecting to an external service or database.

The purpose of an API key is:

- Authentication: It proves to the server which application is making the request.
- Authorization: It tells the server what specific data or features the application is allowed to access.
- Rate Limiting: It tracks how many requests your app makes

so the service can prevent overload or manage billing.

How is it stored and protected:

- Use Environment Variables: Keep keys in separate configuration files or system environment variables, not in your code.
- Use Secret Managers: Store keys in secure cloud tools like AWS Secret Manager or GitHub Secrets.
- Use .gitignore Files: Add your key files to ignore lists so you never accidentally commit and push sensitive, private, or unnecessary files to GitHub.

If an API key is exposed publicly there will be:

- Financial loss: Hackers can still your key to run expensive services or AI models on your dime.
- Data breaches: Attackers can read, change, or delete private user data.
- Service bans: If an thief abuses your key, the service provider might shut down your entire account.

21. A REST API is an architectural style for building web services that allows different software applications to communicate with each other over the internet. REST stands for Representational State Transfer. It relies on a stateless, client-server model where data and functionality are treated as "resources" and accessed using URL.

The main HTTP methods used in REST APIs are:

- GET (Read): This method is used to retrieve data from the server.
- POST (Create): This method is used to send data to the server to create a new resource.
- PUT (Update/Replace): This method is used to update or

completely replace one existing resource.

- DELETE (Delete): This method is used to remove a specific resource from the server.

2. JSON, which stands for JavaScript Object Notation, is a lightweight, text-based format used for storing and transporting data. Although it originates from JavaScript, it is completely language-independent, meaning almost all modern programming languages can read and generate it.

JSON has become the standard for data exchange on the web for several key reasons:

- Human-Readable: It uses a simple, clean, syntax that is easy for humans to read, write, and debug compared to older formats like XML.
- Lightweight: JSON has minimal formatting overhead. This keeps file sizes small, which leads to faster data transmission over the internet.
- Easy for computers to Parse: Web browsers and programming languages can instantly convert JSON text into native objects or data structures with built-in functions.
- Structured format: It naturally supports complex, nested data by organizing information into simple pairs of names (keys) and values, as well as ordered lists (of arrays)

A simple JSON object

```
{  
  "name": "Thabo",  
  "age": 25,  
  "city": "Soweto",  
  "isStudent": false  
}
```

22. Prompt Engineering is the practice of structuring, refining, and designing text inputs (prompts) to guide Generative AI models (like ChatGPT or Claude) to produce the most accurate, relevant, and high-quality outputs. Think of it as the art of learning how to ask the right questions in the right way to get the best possible results.

Prompt Engineering is a crucial skill, because while modern AI tools are incredibly capable, they operate based on patterns and probabilities rather than true human intuition. Good prompt engineering is essential because it: Eliminates ambiguity, saves time and resources, unlocks advanced capabilities.

Two examples of poorly written prompt and their improved versions:

- Poor prompt: "Write an email asking for a deadline extension."
- Improved version: "Can you write a polite, professional email to my project manager, Sarah? Explain that our team needs a 3 day extension on the Q3 Marketing Report because our data analyst was out sick this week. Keep it under 150 words and end by proposing a specific new submission date of next Thursday."
- Poor prompt: "write code to filter an array of users."

Improved version: Using JavaScript (ES6+), write a clean, optimized function called `getPremiumUsers`. The function should take an array of user objects and return only the users who have membership level: "premium" and are is Active: true. Please include a short code comment explaining how it works.

24. A map function is a built-in programming tool that applies a specific operation to every item in a list or collection and returns the new, transformed results.

What it does is to:

- Transform data: It takes each element from an input collection like a list ~~or~~ array, runs it through a function, and collects the output.
- Replace Loops: It lets you avoid writing long while loops to update items one by one.
- Keeps original size: The output collection has the exact same number of items as the input collection.

Python

```
numbers = [1, 2, 3, 4]
```

```
doubled = list(map(lambda x: x * 2, numbers))
```

25. Cloud computing is the delivery of computing services including servers, storage, databases, networking, software, and analytics over the internet ("the cloud"). Instead of buying and maintaining physical data centers and hard drives, companies rent access to these resources from cloud providers on a pay-as-you-go basis.

The three main cloud service models are:

- Infrastructure as a Service (IaaS): The cloud provider rents out basic building blocks like virtual servers, storage space, and networking hardware. You are entirely responsible for installing and managing the operating system, database software, and applications.

Examples: Amazon EC2 or Microsoft Azure Virtual Machines.

- Platform as a Service (PaaS): It provides a pre-configured cloud environment tailored for building, testing, deploying, and managing applications. The provider manages the underlying servers, operating systems, storage, and networking. Developers only need to focus on writing code and managing their applications, completely bypassing server configuration.
Example: Heroku or Google App Engine

Software as a Service (SaaS): Delivers a fully functioning, ready-to-use software application over the internet via a web, browser or app. The vendor completely handles the infrastructure, platform, security, and updates. You simply log in and use it, usually through a monthly or annual subscription model.
Example: Google Workspace (Gmail / Docs) or Microsoft 365.

26. A Version Control System (VCS) is a software tool that tracks and manages changes to files or source code over time. It maintains a detailed history of all modifications, allowing users to review earlier versions, undo mistakes, and collaborate seamlessly without overwriting each other's work.

Git is a free, open-source, distributed version control system created by Linus Torvalds in 2005. Unlike older systems that rely on a single central server, Git is "distributed" because every user downloads a complete copy of the project's entire history onto their local machine.

It is widely used by analysts and developers because it allows:

- Safe experimentation, seamless collaboration, error recovery and Audit Trail and accountability.

- A commit is a saved snapshot of your project's files at a specific point in time.

- A branch is an independent, parallel line of development. When you want to work on a new feature or try out a different data visualization technique you "branch off" from the main project.

27. A Jupyter Notebook is a free, open-source web application that allows you to create and share computational documents containing live code, interactive visualizations, equations and explanatory narrative text.

- While Jupyter Notebook supports over 40 programming languages through different language engines called "kernels", it is most commonly used with Python.

Jupyter Notebook has become the de facto standard for communication and experimentation in data science and AI due to several unique features:

- Cell-by-cell Execution: Instead of running a massive script all at once, analysts can isolate and execute single blocks of code. This makes data cleaning, debugging, and testing hypothesis incredibly fast and interactive.

- Immediate Feedback & Rich Outputs: Result data tables, and visualizations (using libraries like Matplotlib or Plotly) render directly underneath the code cell that generated them. This allows data professionals to observe the immediate effect of their data manipulation.

- Persistent Memory State: When a code cell is executed, the variables and objects remain stored in the memory session.

(the kernel). Data scientists can load a massive dataset once and repeatedly experiment on it without wasting time reloading the data.

- "Literate Programming" (Documentation): Jupyter blends code with Markdown, images, and LaTeX mathematical equations. Instead of just turning in code, a data scientist can build a complete, readable story explaining why they chose a certain approach.
- Easy Sharing and Reproducibility: Notebooks can be saved as .ipynb files or natively exported into standard presentation formats like HTML, PDF, or slides. This makes it simple to share reproducible workflows with peer reviews or business stakeholders.

28. Python is a high-level, user friendly language that is interpreted line by line, which makes it easy to read and write. Because it is versatile and general-purpose, it can be applied to almost any task, including web development, software creation, task automation, and building machine learning models.

Python is considered the most popular programming language for Data analytics and AI because of:

- Readability and Simplicity: Python uses clean syntax that closely resembles plain English. This allows data professionals to focus on solving complex data problems rather than fighting with complicated code structure.

Massive Ecosystem of Libraries: Instead of writing complex mathematical formulas or machine learning algorithms from scratch, developers can leverage thousands of pre-built packages tailored specifically for data work.

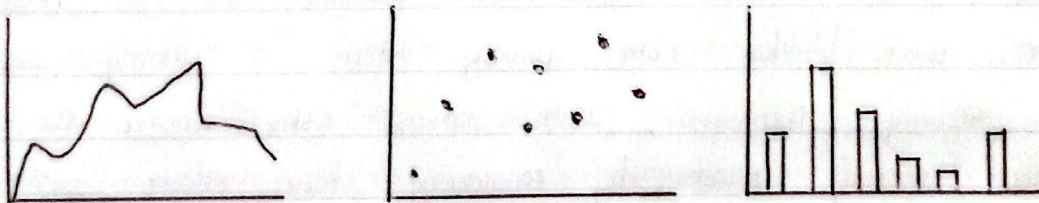
Strong community and support: Python has one of the

largest developer communities in the world. This ensures continuous updates, extensive documentation, and a wealth of tutorials for trouble shooting.

- Seamless Integration: It easily connects with other data tools, cloud platforms, and database languages (like SQL), making it a perfect fit for modern data engineering pipelines.

Four Python libraries commonly used in data work are:

- Pandas: The ultimate tool for data manipulation and analysis. It introduces the "DataFrame" (a powerful, tabular data structure similar to an Excel spreadsheet) to easily clean, filter, and restructure raw data.
- Numpy: Short for Numerical Python, this library provides support for large, multi-dimensional arrays and matrices, along with a massive collection of high-level mathematical functions to operate on them efficiently.
- Matplotlib: A foundational library for data visualization. It allows users to generate static, animated, and interactive charts such as line graphs, bar charts, and scatter plots directly from the data.



- Scikit-learn: It features simple and efficient tools for predictive data analysis, including algorithms for classification, regression, clustering, and model evaluation.

Section D: Data Ethics, Governance & Careers

29. Data privacy is the right of individuals to decide how their personal information is collected, used and shared.

Data privacy is important because it:

- Prevents identity theft: stops criminals from stealing and misusing personal data.
- Builds trust: Helps customers feel safe when sharing details with businesses.
- Gives control: Lets people see, correct, or delete their stored information.
- Avoids penalties: Keeps companies safe from large fines for data leaks.

The General Data Protection Regulation (GDPR) is a strict privacy law created by the European Union (EU). It took effect on May 25, 2018. It gives citizens more power over their digital footprint and forces companies to be transparent about data collection.

It applies to all countries that are members of the European Union (EU) and the European Economic Area (EEA).

30. Data ethics looks at the right and wrong of how we handle data, from how it is collected and protected to how it is shared and used. Unlike data privacy which is mostly about laws and security measures, data ethics asks whether our actions are morally correct, by considering their effects on people's rights, fairness, freedom, and equality.

Here are three major examples of the unethical use of data or AI, along with why they are highly problematic.

1. Algorithm Bias and Automated Hiring.

Example: A company builds an AI tool to screen thousands of job CVs. To train the AI, developers feed it historical employee data from the past ten years. Because the industry historically hired mostly men, the AI teaches itself that male candidates are preferable and systematically downgrades CVs containing words like "women's" (e.g. "women's chess club captain").

Why it is problematic: It perpetuates systemic discrimination. AI is often viewed as a neutral decision-maker, but it acts as a mirror to our past societal biases. When biased historical data is automated, it robs qualified individuals of equal career opportunities while masking discrimination under the guise of "objective" software metrics.

2. Political Microtargeting Without Consent (Surveillance Capitalism):

Example: A data firm surreptitiously scrapes the private profile data, personality traits, and "likes" of tens of millions of social media users without their explicit knowledge. They then use this behavioral data to feed users highly manipulative, psychologically tailored political ads designed to inflame fear.

Why it is problematic: It erodes human autonomy and democratic processes. Instead of persuading users through transparent public debate, this practice weaponises cognitive vulnerabilities using stolen information. It bypasses informed consent, treats human behavior strictly as a commodity, and polarises societies by trapping people inside algorithmic echo chambers.

3. Racial Disparities in AI Facial Recognition & Predictive Policing

Example: Law enforcement agencies deploy facial recognition software to identify suspects from public CCTV cameras.

or use predictive software to dispatch officers to "high-risk" neighborhoods. However, the software was trained on datasets that heavily lacked diverse skin tones, leading to drastically higher error rates for people of color.

Why it is problematic: It threatens human rights and basic liberties, leading to catastrophic real-world errors. This specific ethical failure has directly resulted in the wrongful arrest and humiliation of innocent people due to "false-matches".

31. Data governance is the formal system of rules, processes, and responsibilities that an organization uses to manage its data assets. It defines who has control and authority over data, how data can be used, and how its security, accuracy, and availability are maintained throughout its lifecycle. Think of it as the construction that dictates how data is handled across a business.

The key components of a Data Governance framework are:

1. Strategic Alignment & Objectives

- Data strategy: a clear roadmap aligning data initiatives with overall business goals (e.g. improving customer retention or complying with POPIA)
- Key Performance Indicators (KPIs): Measurable metrics to track the success and maturity of data initiatives.

2. Organizational Roles & Responsibilities

- Data Governance Council: A steering committee of executive leaders that sets policies, resolves disputes, and funds data projects.
- Data Stewards: Tactical experts responsible for the day-to-day

- quality and maintenance of specific data domains.
- Data Owners: Senior managers accountable for the data within their specific business unit.

3. Data Policies & Standards

- Data Lifecycle Management: Documented rules outlining exactly how data is captured, stored, archived, and permanently destroyed.
- Compliance Rules: Policies that align operations with legal frameworks like the GDPR or POPIA.
- Access Control: Definitive rules stating who is allowed to view, modify, or share specific data tiers.

4. Data Quality & Metadata Management

- Data Quality Frameworks: Automated checks to ensure data is accurate, complete, consistent and up to date.
- Metadata Management: Creating a "data about data" directory (or data catalog) so employees can easily search for, find, and understand.
- Data Lineage: Mapping the visual journey of data to see where it originated, how it moved, and how it changed over time.

5. Security & Privacy Controls

- Data Classification: Categorizing data based on sensitivity (eg Public, Internal, Confidential, Restricted)
- Anonymization & Masking: Masking sensitive data fields (like ID numbers or banking details) so unauthorized users or developers cannot see them.

32. Bias in AI refers to a situation where an AI model produces systematically skewed, unfair, or prejudiced results. It occurs when an AI algorithm reflects human biases, systemic inequalities, or flaws introduced during its development, training, or deployment.

A Real-World Example: Amazon's Biased Hiring Tool

In the mid-2010s, Amazon developed an experimental AI recruiting tool designed to review job applications and score applicants from one to five stars. The goal was to automate the screening process and find top talent efficiently.

However, because the tech industry was historically male-dominated, the AI model was trained on a dataset consisting of CVs submitted over a 10-year period, most of which came from men.

Why it was biased: The algorithm taught itself that male candidates were preferable. It began actively penalising applicants that included the word "women's" (such as "women's rugby club captain") and downgraded graduates from two specific all-women's colleges.

Potential Consequences of AI Bias

- Socioeconomic Exclusion: Qualified individuals can be locked out of life-changing opportunities.
- Systemic Injustice and Harm: In criminal justice, biased facial recognition software has led to the wrongful arrests of individuals.
- Erosion of Public Trust: When automated systems consistently make unfair or harmful errors, the public loses trust in both

the technology and the institution using it.

- Legal and Financial Penalties: Companies utilizing biased AI risk severe reputational damage, expensive class-action lawsuits, and heavy regulatory fines under modern frameworks like the EU AI Act.

33. While all four roles work with data, they focus on different stages of the data lifecycle. A Data Engineer builds the infrastructure, a Data Analyst interprets historical data, a Data Scientist builds predictive models, and an AI Engineer integrates those models into scalable software applications.

Job Title	Primary Role	Key Skills	Everyday Tools
Data Analyst	Interprets historical data to help businesses make informed decisions and track performance metrics	Data cleaning, data visualization, business intelligence, A/B testing, and storytelling.	SQL, Excel, Tableau, Microsoft Power BI
Data Scientist	Build predictive models and discover hidden patterns in data.	Advanced statistics, machine learning, Python.	Python, R, Scikit-learn, Jupyter.
Data Engineer	Build and maintain the infrastructure that stores and moves data.	Database design, ETL (Extract, Transform, Load), cloud computing	Apache, Airflow, Kafka, Spark, SQL.
AI Engineer	Create, test, and run smart systems and neural networks.	Deep learning, natural language processing, software deployment.	PyTorch, TensorFlow, Python, cloud platforms

24. Business Intelligence (BI) refers to the technology, processes, and strategies used to turn raw corporate data into actionable insights for decision-making.

The two popular BI tools are;

Power BI - it allows companies to aggregate data from disparate sources (like Excel sheets, SQL databases, and cloud services) into unified, interactive dashboards. Decision-makers use these real-time charts to track key performance indicators (KPI) and immediately spot operational bottlenecks or sales trends.

Tableau - it excels at handling massive datasets and transforming complex data into intuitive, drag-and-drop visuals. Teams can conduct ad-hoc analyses and drill down into specific data points on the fly, enabling executives to rapidly validate business hypotheses and make strategic shifts based on live visual evidence.

SECTION E: Personal Reflection

25. Computer Vision (CV) is the concept that I find to be the most interesting in this assignment.

Just the whole idea of computers being able to read, interpret and respond to images make me realise how advanced technology has become. It also makes me realise that the human mind can achieve anything they put their mind to; because its humans who came up with this kind of technology.

26. Training Dataset is one concept that I found difficult to understand. I used Google and then copied the information to ChatGPT and requested for a simpler explanation of the concept.

37. I think the modern world is moving towards artificial intelligence at a rapid speed, and I personally believe that in my career path which is technical, things are looking good, there will be more job opportunities for those who are technical. However I cannot say the same for those with non-technical careers, some might end up losing their jobs since AI is taking over.

Eg jobs that usually needed human beings to perform them are being computerised. In retail stores there won't be a need for cashiers in the future because there is a technology that allows people (customers) to do everything themselves. This technology has not been implemented in our country yet but it probably will, in the near future.

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